

MACHINE LEARNING HOME WORK - 3

① K-means clustering (Manhattan Distance, $K=2$)

Data points : $(1, 2), (2, 2), (4, 4), (5, 4), (5, 5)$

Initial centroids : $C_1 = (1, 2)$

$C_2 = (2, 2)$

Step 1 : Cluster Assignment using Manhattan distance.

Manhattan distance b/w points (x_1, y_1) & (x_2, y_2)
 $= |x_1 - x_2| + |y_1 - y_2|$

Given datapoint	dist. to $C_1 = (1, 2)$	dist. to $C_2 = (2, 2)$	Assigned Cluster
$(1, 2)$	0	1	C_1
$(2, 2)$	1	0	C_2
$(4, 4)$	5	4	C_2
$(5, 4)$	6	5	C_2
$(5, 5)$	7	6	C_2

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Cluster assignment after step 1.

$$C_1: (1, 2)$$

$$C_2: (2, 2), (4, 4), (5, 4), (5, 5)$$

Step 2: Computing new centroids

$$C_1 \text{ mean} = (1, 2)$$

$$C_2 \text{ mean} = \text{median of } x: [2, 4, 5, 5] = 4.5$$

$$\text{median of } y: [2, 4, 4, 5] = 4$$

$$\text{New } C_2 = (4.5, 4)$$

Step 3: Reassign clusters with using centroids

$$C_1 = (1, 2), C_2 = (4.5, 4)$$

Point	Dist. to $C_1 (1, 2)$	Dist. to $C_2 (4.5, 4)$	Assigned cluster
(1, 2)	0	6.5	C_1
(2, 2)	1	3.5	C_1
(4, 4)	5	0.5	C_2
(5, 4)	6	1.5	C_2
(5, 5)	7	2.5	C_2

New Clusters Assignment:

$$C_1 : (1,2), (2,2)$$

$$C_2 : (4,4), (5,4), (5,5)$$

Step 4: Compute new medians again.

for $C_1 = (1,2), (2,2)$

$$x = [1,2] \rightarrow \text{median} = 1.5$$

$$y = [2,2] \rightarrow \text{median} = 2$$

$$\rightarrow \text{new } C_1 = (1.5, 2.0)$$

for $C_2 = (4,4), (5,4), (5,5)$

$$x = [4,5,5] \rightarrow \text{median} = 5$$

$$y = [4,4,5] \rightarrow \text{median} = 4$$

$$\rightarrow \text{new } C_2 = (5, 4.0)$$

Step 5: Repeating Step 3.

Points	dist. to C_1	distance to C_2	Assigned cluster
(1,2)	0.5	6	C_1
(2,2)	0.5	5	C_1
(4,4)	4.5	1	C_2
(5,4)	5.5	0	C_2
(5,5)	6.5	1	C_2

New cluster assignment :

$$C_1 = (1, 2), (2, 2)$$

$$C_2 = (4, 4), (5, 4), (5, 5).$$

Cluster assignment remains same, showing convergence and the algorithm stops.

② ~~entropy~~ of the g

Step 1: Calculate entropy of the root

we have :

6 high

2 low

$$\text{Entropy}(S) = -P^+ \log_2(P^+) - P^- \log_2(P^-)$$

$$= -\frac{6}{8} \log_2 \frac{6}{8} - \frac{2}{8} \log_2 \frac{2}{8}$$

$$= -0.75 \log_2(0.75) - 0.25 \log_2(0.25)$$

$$\approx 0.811$$

Step 2: Information gain for each attribute.

< 25 : 5 samples : 5 high, 0 low $\rightarrow$ entropy = 0

≥ 25 : 3 samples $\rightarrow$ 1 high, 2 low

Attribute : Age

< 25 : 5 Samples $\rightarrow$ 5 High, 0 low $\rightarrow$ Entropy = 0

≥ 25 : 3 Samples $\rightarrow$ 1 High, 2 low

$$\text{Entropy} = -\frac{1}{3} \log_2 \frac{1}{3} - \frac{2}{3} \log_2 \frac{2}{3} \approx 0.918$$

weighted entropy :

$$\frac{5}{8} (0) + \frac{3}{8} (0.918) \approx 0.344$$

$$\text{Gain (Age)} = 0.8111 - 0.344$$

$$= 0.467$$

Attribute : income

$> 50K$: 4 Samples : 2 high, 2 low

$< 50K$: 4 Samples : all high.

$$\text{entropy} = -\frac{2}{4} \log_2 \left(\frac{2}{4} \right) - \frac{2}{4} \log_2 \left(\frac{2}{4} \right)$$

$$= -0.5 \cdot 1 - 0.5 \cdot 1$$

$$= 1$$

gain (income) = entropy (S) - entropy split.

$$\text{entropy split} = \frac{4}{8} \cdot 1 + \frac{4}{8} \cdot 0$$

$$= 0.5$$

$$\text{gain (income)} = 0.8111 - 0.5$$

$$= 0.311$$

Attribute : gender

M: 4 samples $\rightarrow$ 3 high, 1 low $\rightarrow$ entropy ≈ 0.811

F: 4 samples $\rightarrow$ 3 high, 1 low $\rightarrow$ entropy ≈ 0.811

weighted entropy = 0.811

Gain (gender) = $0.811 - 0.811$

= 0

Hence, we choose age to split.

Step 3. Build the tree.

Root : Age.

If $< 25 \rightarrow$ 5 samples, all high $\rightarrow$ leaf = high

If $\geq 25 \rightarrow$ 3 samples

Income

Gender

Risk

< 50K

F

High

> 50K

F

Low

> 50K

M

Low

1 high, 2 low $\rightarrow$ entropy = 0.918

Subtree for Age ≥ 25

Information gain on this subset:

Income:

$< 50K$: 1 high $\rightarrow$ entropy = 0

$> 50K$: 2 low $\rightarrow$ entropy = 0

$$\text{gain} = 0.918 - 0 = 0.918$$

Gender:

F: 1 high, 1 low $\rightarrow$ entropy = 1

M: 1 low $\rightarrow$ entropy = 0

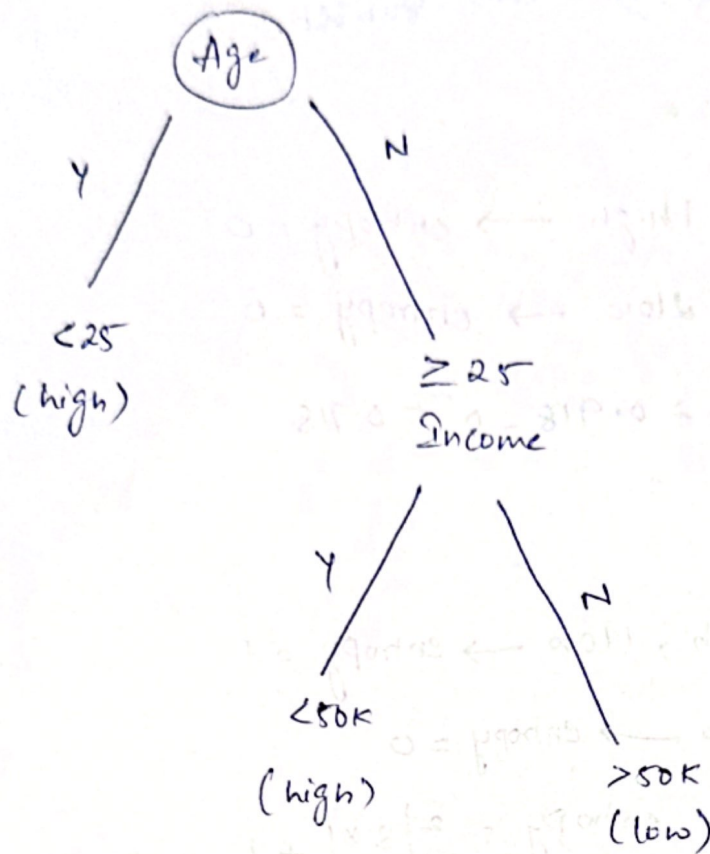
$$\text{weighted entropy} = \frac{2}{3} \times 1 + \frac{1}{3} \times 0$$

$$= 0.667$$

$$\text{gain} = 0.918 - 0.667 = 0.251$$

Hence the 2nd decision node for right child node is income.

Final decision tree.



4.

- Suppose the number of A, C, G or T is a, b, c, d respectively.

$$\text{Likelihood} : P(a, b, c, d | \theta) = c \left(\frac{1}{3}\right)^a \theta^b \left(\frac{1}{3}\right)^c \left(\frac{1}{3} - \theta\right)^d$$

$\approx$

$$\ln P(a, b, c, d | \theta) = \ln c + a \ln \frac{1}{3} + b \ln \theta + c \ln \frac{1}{3} + d \ln \left(\frac{1}{3} - \theta\right).$$

normalizing the likelihood

(derivative = 0)

$$\frac{b}{b} = \frac{d}{\frac{1}{3} - \theta} = 20$$

$$\theta = \frac{b}{3(b+d)}$$

- expectation - maximization (EM) Algorithm

E-Step. (expectation step).

if we know the value θ , we can estimate the expected value of b & d :

$$\frac{a}{b} = \frac{1}{30}$$

$$\frac{a+b}{b} = \frac{1+30}{30}$$

$$\frac{x}{b} = \frac{1+30}{30}$$

$$b = \frac{30}{1+30} x$$

$$a = x - b$$

$$\frac{c}{d} = \frac{1}{1-30} \Rightarrow \frac{c+d}{d} = \frac{2-30}{1-30} \Rightarrow \frac{y}{d} = \frac{2-30}{1-30}$$

$$d = \frac{1-30}{2-30} y$$

$$c = y - d$$

M-step (maximization step)

we know b & d , then we can find θ maximize the likelihood.

$$\theta = \frac{b}{3(b+d)}$$

⑤ Linear SVM Calculation.

given,

only three non-zero Lagrange multipliers (support vectors).

i	α_i	x_i	y_i
1	0.05	$[0, 2]$	-1
2	0.05	$[2, 1]$	-1
3	0.10	$[3, 5.5]$	1

Step 1: compute weight vector w

Formula:

$$w = \sum_i \alpha_i y_i x_i$$

$$w = 0.05 \cdot (-1) \cdot [0, 2] + 0.05 \cdot (-1) \cdot [2, 1] + 0.10 \cdot (1) \cdot [3, 5.5]$$

$$[3, 5.5]$$

Break it down :

$$0.05 \cdot -1 \cdot [0, 2] = [0, -0.1]$$

$$0.05 \cdot -1 \cdot [2, 1] = [-0.1, -0.05]$$

$$0.10 \cdot 1 \cdot [3, 5.5] = [0.3, 0.55]$$

Add them :

$$w = [0 + (-0.1) + 0.3, -0.1 + (-0.05) + 0.55]$$

$$w = [0.2, 0.4]$$

Step 2 : compute Bias b .

use any support vector, Let's use $x_1 = [0, 2], y_1 = -1$

formula:

$$y_i(w \cdot x_i + b) = 1 \Rightarrow -1 \cdot (0.2 \cdot 0 + 0.4 \cdot 2 + b) = 1$$

$$-1 \cdot (0.8 + b) = 1 \Rightarrow 0.8 - b = 1 \Rightarrow b = -1.8$$

Step 3 : compute Margin

$$\text{Margin} = \frac{1}{\|w\|} = \frac{1}{\sqrt{0.2^2 + 0.4^2}} = \frac{1}{\sqrt{0.04 + 0.16}} = \frac{1}{\sqrt{0.2}}$$

$$= \frac{1}{0.447} \approx 2.236$$

$$\text{Margin} \approx 2.236$$